Lea Yanhui Li

AI Safety, Evaluation & Risk Infrastructure · Sr. Staff TPM / PM

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Sr. Staff TPM specializing in AI safety, evaluation, and risk infrastructure at scale, with 15+ years translating policy and technical evidence into production decisions where failure carries regulatory, reputational, or human risk. At Meta, owned reliability of ML classifier pipelines and LLM-powered enforcement systems — purpose limitation enforcement, evaluation-driven deployment, human-in-the-loop review, audit trails, and operational oversight for safety-critical automation across millions of assets and data flows. Independently built an LLM safety evaluation and release gating system — multi-turn trajectory risk tracking, irreversibility-aware action gating and structured SHIP / CONDITIONAL SHIP / BLOCK decisions on policy compliance.

CORE COMPETENCIES

Safety & Evaluation: ML classifier reliability · LLM-as-judge design · evaluation infrastructure · release gating · trajectory risk accumulation · irreversibility-aware action gates · ground-truth datasets · adversarial datasets · refusal quality scoring · failure-mode analysis · SLO definition & tracking · drift detection · post-mortem execution

Program Leadership: 0-to-1 program ownership · operational excellence · cross-org alignment · influence without authority · risk-weighted decisions · policy operationalization · runbook development · executive stakeholder communication · multi-year roadmaps

Technical: Python · Anthropic / OpenAI APIs · CI/CD · AWS (SA Certified) · GCP · Kubernetes · LLM safety evaluation · ML evaluation infrastructure · data lineage pipelines · adversarial dataset design

Domains: AI governance · safety systems · agentic safety · responsible AI deployment · regulatory compliance · privacy

EXPERIENCE

Sr. Staff Technical Program Manager · Risk & Privacy Infrastructure · *Meta · Menlo Park, CA* 2022 – 2026

Owned strategy and execution for Meta's Privacy-Aware Infrastructure — enforcing that user data is used only for its stated purpose across Instagram, Facebook, MSL, Fintech, and Reality Labs. Focus: how ML classifiers and LLM-powered systems are evaluated, deployed, and monitored where mis-calibrated automation means compliance breach or harm at scale.

- **Cross-org program ownership:** Built a 0→1 ML automation program across 50+ engineers and 5 product orgs with no shared OKR, delivering **$110M annual cost reduction** while maintaining all enforcement SLOs and regulatory guardrails.
- **Evaluation infrastructure & classifier reliability:** Designed evaluation-driven deployment systems for ML classifiers — automation threshold design, drift detection, and risk-tiered precision/recall calibration against ground-truth datasets. Improved precision **80% → 95%** across safety-critical pipelines.
- **Operational reliability & incident response:** Built observability infrastructure for LLM-powered enforcement — regression monitoring, post-mortem execution, human-in-the-loop review workflows with defined SLOs and runbooks.
- **Purpose limitation & harm prevention at scale:** Redesigned fragmented manual checks into end-to-end enforcement flows — ML-powered data discovery, lineage traversal, consent verification, and automated evidencing — reducing cycle time from **17 days to 5 minutes**.
- **Youth safety, fintech & sensitive data:** Led age discovery and enforcement for Teen Account launches (EU + Global), reducing age misuse by **95%**; deployed PCI-compliant payment data traceability across 100% of Meta Fintech cloud-to-on-premise endpoints, uncovering and remediating 8+ sensitive data leaks.

Senior Technical Program Manager · Amazon Chime SDK · *Amazon Web Services · Santa Clara, CA* 2021 – 2022

Bridged research science and production for AI-powered audio and video features. Partnered with research scientists and applied scientists across the full pipeline — from training data collection through model evaluation, launch readiness, and feature delivery. Delivered Voice Focus and Background Blur, improving real-time communication quality at scale.

AI SAFETY & EVALUATION PORTFOLIO

LLM Safety Evaluation & Release Gating System · GitHub

- Translates written safety policy (YAML) into measurable evaluation categories, risk-weighted thresholds, and structured SHIP / CONDITIONAL SHIP / BLOCK decisions. LLM-as-judge scoring across adversarial and indirect attacks assesses refusal quality, harmfulness, and policy compliance —penalizing cold or dismissive refusals on critical-severity prompts.
- **Phase 1 Trajectory risk accumulator:** Extended single-turn gating to multi-turn conversation tracking with weighted risk accumulation across turns. Each turn evaluated independently; trajectory score escalates non-linearly when later turns reveal

harmful intent that no individual turn would trigger. Addresses the multi-turn harm prevention gap that single-turn classifiers are structurally blind to.

- **Phase 2 Irreversibility-aware action gate:** Added tool-call evaluation as a separate dimension from text outputs — scoring proposed actions on harm probability AND reversibility. Irreversible actions (send_email, file_delete, external API calls with side effects) require higher confidence to SHIP and trigger immediate BLOCK in suspicious trajectories. Implements the threshold logic that distinguishes inference safety from agentic safety.
- **Key finding:** evaluated GPT-4o across 62 adversarial prompts — heuristic vs. LLM-as-judge evaluation produced **85.7% vs. 28.6% self-harm failure rate** (3× difference). Demonstrates that evaluation method choice is itself a safety design decision with measurable consequences. Revealed a consistent bypass pattern across indirect framings — fiction, academic pretext, sentimental framing — with failure rates of 28.6% (self-harm), 35.7% (illicit behavior), 16.7% (jailbreak).

LLM Evaluation Framework · GitHub

CI/CD-integrated benchmarking system measuring hallucination rate, classification accuracy, refusal compliance, and consistency across model versions. Caught GPT-4o fabricating a non-existent Einstein lecture (Claude Haiku 4.5: 100% vs GPT-4o: 91.7%).

EARLIER EXPERIENCE

Senior Research Scientist & Enterprise Architect · *Hitachi America* 2016 – 2019

Led AI-driven IoT analytics and anomaly detection for smart manufacturing; improved productivity 17%.

Co-Founder & Chief Product Officer · *SuiteSocial* 2019

Built a 0-to-1 ML-driven marketplace matching brands and influencers; onboarded 300+ influencers and 8 paying clients.

Director of Engineering & Product (part-time) · *CommunityConnect Labs* 2019 – 2020

Led product and engineering for a conversational AI system serving government clients; managed a cross-continental team.

Software Engineer · *Oracle* · *Motorola* 2008 – 2016

Developed core components of Oracle's Virtual Desktop Infrastructure platform supporting enterprise virtualization and cloud deployments; shipped two mobile products at Motorola.

EDUCATION

MBA, University of California, Berkeley, Haas School of Business

MEng, Computer Engineering, National University of Singapore

BEng, Computer Science, Northeastern University, China

CERTIFICATIONS & LEADERSHIP

- Google Generative AI Leader — Google (2026)
- Advanced PM Skills — Product Faculty (2025)
- AI Product Management Certificate — Product Faculty (2024)
- AWS Certified Solutions Architect – Associate (2021)
- PMP — Project Management Professional (2022)
- Mentor, Berkeley Haas Entrepreneurship Program (2024 – Present)
- Lecturer, Operations Management — UC Berkeley Haas (2021)